

SOCIOL 723: Social Statistics II

Week 7, Instrumental Variables

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Today

Endogeneity and the IV Estimator

Two Canonical Applications

Two-Stage Least Squares

What Does IV Estimate?

Weak Instruments and Inference

Practice

A Change of Strategy

Week 6 assumed we had measured every confounder and worked on *how* to adjust. That assumption is usually false, and no diagnostic can rescue it.

The instrumental variables idea

Give up on controlling for confounders. Instead, find a source of variation in the treatment that is *as good as randomly assigned*, and use only that variation.

- We throw away most of the variation in D and keep a sliver we can defend. Precision falls; credibility rises.
- The price is steep and often hidden: we no longer estimate the ATE, and the assumptions are as untestable as ignorability was, just different, and usually easier to argue about in substantive terms.

Endogeneity

What Goes Wrong, in Scalar Form

We want β in $Y_i = \alpha + \beta D_i + \epsilon_i$, but D_i is **endogenous**: $\text{Cov}(D_i, \epsilon_i) \neq 0$. From Week 1,

$$\hat{\beta} = \frac{\text{Cov}_n(D_i, Y_i)}{\text{Var}_n(D_i)} \xrightarrow{p} \beta + \frac{\text{Cov}(D_i, \epsilon_i)}{\text{Var}(D_i)}.$$

The bias term does not shrink with n : OLS is *inconsistent*, not merely biased. More data buys precision around the wrong number.

- **Omitted variables**: unmeasured ability, motivation, family resources. This is the OVB formula from Week 1, restated.
- **Simultaneity**: Y also causes D , police and crime, aid and growth, marriage and earnings.
- **Measurement error in D** : classical error attenuates $\hat{\beta}$ toward zero, by a factor equal to the reliability ratio.
- **Selection** into the sample on a collider (Week 5).

Sign and Size of the Bias

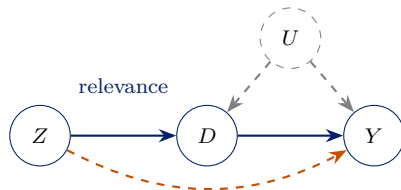
It is worth being concrete about what we are trying to remove. Writing $\epsilon_i = \gamma A_i + u_i$ with A_i an unmeasured confounder,

$$\text{plim } \hat{\beta}_{OLS} = \beta + \gamma \frac{\text{Cov}(D_i, A_i)}{\text{Var}(D_i)}.$$

- For schooling and earnings: ability raises earnings ($\gamma > 0$) and is positively selected into schooling, so OLS is biased *up*.
- Measurement error in reported schooling pushes the other way. The two biases are of unknown relative size, which is why the sign of the total OLS bias is genuinely uncertain, and why the empirical finding that IV estimates *exceed* OLS is not automatically evidence of a bad instrument.
- Before running IV, say which bias you are trying to remove and in which direction it runs. If you cannot, the exercise has no target.

An Instrument (cont.)

An instrument Z_i moves D_i but is otherwise unrelated to Y_i .



must not exist

An Instrument (cont.)

Three conditions, stated carefully

1. **Relevance:** $\text{Cov}(Z_i, D_i) \neq 0$. **Testable.**
2. **Independence:** Z_i is as good as randomly assigned, no arrow into Z from anything that also causes Y .
3. **Exclusion:** Z affects Y *only* through D . **Not testable.**

Conditions 2 and 3 are distinct and are often conflated. A randomized lottery satisfies 2 by construction and can still violate 3.

The Wald Estimator

Take covariances of $Y_i = \alpha + \beta D_i + \epsilon_i$ with Z_i :

$$\text{Cov}(Z_i, Y_i) = \beta \text{Cov}(Z_i, D_i) + \underbrace{\text{Cov}(Z_i, \epsilon_i)}_{=0} \implies \boxed{\beta_{IV} = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, D_i)}}$$

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With a *binary* instrument this is a ratio of two differences in means:

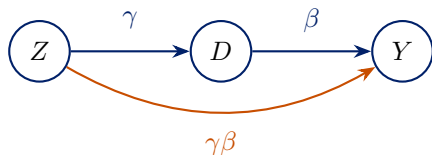
$$\beta_{IV} = \frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]} = \frac{\text{reduced form}}{\text{first stage}}.$$

Read it as: *the effect of the instrument on the outcome, scaled up by how much the instrument moved the treatment.*

Intuition for the Ratio

Divide numerator and denominator by $\text{Var}(Z_i)$:

$$\beta_{IV} = \frac{\text{Cov}(Z_i, Y_i) / \text{Var}(Z_i)}{\text{Cov}(Z_i, D_i) / \text{Var}(Z_i)} = \frac{\text{coefficient from regressing } Y \text{ on } Z}{\text{coefficient from regressing } D \text{ on } Z}.$$



- The instrument's total effect on Y travels along one path and equals $\gamma\beta$. The first stage recovers γ . Dividing isolates β .
- If γ is small, we are dividing by something close to zero. Hold that thought; it is the whole of the weak-instrument problem.

Applications

Settler Mortality and Institutions

Acemoglu, Johnson, and Robinson (2001) estimate the effect of institutional quality (D , protection from expropriation risk) on GDP per capita (Y).

- Institutions are obviously endogenous to economic performance: rich countries can afford good institutions, and both may be driven by culture, geography, or history.
- **The instrument:** mortality rates faced by European settlers in the colonial period.
- **The mechanism argument:** where settlers could survive, they settled and built inclusive institutions, property rights, rule of law. Where mortality was high, colonization was extractive. Those institutional forms persisted long after the disease environment ceased to matter.
- **Relevance:** strongly supported by the first stage.

Where the AJR Argument Is Attacked

Exclusion requires that historical settler mortality affects income *today* only through institutions.

- **The leading threat is geography.** The disease environment that killed settlers also shapes agricultural productivity, the modern disease burden, and human capital, all of which affect income directly.
- AJR's response is an *assumption*: that the confounding role of geography is adequately captured by a linear term in distance from the equator plus continent dummies.
- That is a functional-form assumption doing identification work, and it is rarely defended. We come back to it briefly in Week 13.
- Subsequent critiques questioned the mortality data itself. **Note the pattern:** exclusion cannot be tested, so the debate becomes a debate about history, measurement, and functional form. That is healthy, and it is what IV papers should invite.

Quarter of Birth and Schooling (cont.)

Angrist and Krueger (1991) estimate the return to schooling using **quarter of birth** as an instrument.

- **The mechanism:** children start school in the year they turn six, but may leave at a fixed *age* (16 or 17). Those born early in the calendar year reach the leaving age having completed less schooling.
- **Relevance:** the first stage is real but small, a fraction of a year of schooling.
- **Exclusion:** birth quarter should not affect earnings except through schooling. Contested, because season of birth correlates with maternal characteristics, health at birth, and relative age in class.

Quarter of Birth and Schooling (cont.)

- **The compliers** are people who would have dropped out at the minimum age. The LATE is therefore a return to schooling *at the bottom of the schooling distribution*, a different quantity from the average return, and arguably the policy-relevant one for compulsory attendance laws.

For now, read *compliers* as “the people whose schooling was actually moved by their birth quarter,” and *LATE* as “the average effect for those people.” Later in the lecture we define the four compliance types properly and show why the IV ratio can only speak for this one group.

The Cautionary Half of Angrist–Krueger (cont.)

The paper is also the standard illustration of what goes wrong with many weak instruments. It previews two objects we formalize in the next two sections: **2SLS**, the machinery for combining several instruments, and **weak identification**, which explains how adding instruments can make an estimate worse rather than sharper.

- To gain precision, the authors interacted quarter of birth with state and year of birth, producing up to 180 instruments.
- Bound, Jaeger, and Baker (1995) showed that with that many weak instruments, 2SLS is biased *toward OLS*, and demonstrated it devastatingly: replacing quarter of birth with *randomly generated* instruments produced similar estimates and standard errors.
- The mechanism is overfitting the first stage. $\hat{D}$ starts to reproduce the endogenous variation we were trying to discard.

The Cautionary Half of Angrist–Krueger (cont.)

- **More instruments is not more information.** Report the just-identified estimate alongside any overidentified one.

Sociological instruments worth knowing: sibling sex composition for fertility, distance to college for schooling, lottery admission for school quality, and the draft lottery for military service.

2SLS

The Two Stages

With covariates and possibly several instruments:

$$\text{Stage 1: } D_i = \pi_0 + \pi_1 Z_i + \pi_2' W_i + v_i \implies \hat{D}_i$$

$$\text{Stage 2: } Y_i = \alpha + \beta \hat{D}_i + \gamma' W_i + u_i$$

- $\hat{D}_i$ is the part of D_i *explained by the instrument*, which by assumption is uncorrelated with ϵ_i . We regress on the clean part and discard the rest.
- **The exogenous covariates W_i must appear in both stages.** Omitting them from the first stage is a common and serious error.
- **Order condition:** at least as many excluded instruments as endogenous regressors. *Exactly identified* means equality; *overidentified* means more.
- **Never run the two stages by hand.** The second-stage standard errors would treat $\hat{D}$ as data rather than as an estimate. Use `ivreg()` or `feols(y ~ w | d ~ z)`.

IV in Matrix Form ★

Start from the moment condition $E[\mathbf{Z}'u] = \mathbf{0}$ with $Y = D\beta_0 + u$. The sample analogue $\mathbf{Z}'(\mathbf{Y} - \mathbf{D}\hat{\beta}) = \mathbf{0}$ has no solution when there are more instruments than regressors, so instead *minimize* the sample moments in a weighted norm:

$$\hat{\beta} = \arg \min_b (\mathbf{Z}'(\mathbf{Y} - \mathbf{D}b))'(\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'(\mathbf{Y} - \mathbf{D}b)).$$

The first-order condition gives

$$\hat{\beta} = \left(\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D} \right)^{-1} \mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{Y}.$$

IV in Matrix Form: The 2SLS Estimator ★

Define the projection matrix $\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$, the same object as Week 1, now projecting onto the instrument space:

$$\hat{\beta}_{2SLS} = (\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}\mathbf{D}'\mathbf{P}_Z\mathbf{Y}$$

Since $\mathbf{P}_Z\mathbf{D}$ is the first-stage fitted value and $\mathbf{P}_Z$ is idempotent, this is literally “regress $\mathbf{Y}$ on $\hat{\mathbf{D}}$.” And when $\mathbf{Z} = \mathbf{D}$, $\mathbf{P}_Z\mathbf{D} = \mathbf{D}$ and 2SLS collapses to OLS: OLS is the special case in which you claim everything is exogenous.

The IV Sandwich ★

Substitute $\mathbf{Y} = \mathbf{D}\beta_0 + \mathbf{u}$:

$$\hat{\beta} - \beta_0 = (\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}\mathbf{D}'\mathbf{P}_Z\mathbf{u}.$$

Multiply by $\sqrt{n}$. Under regularity conditions and a *strong* instrument,

$$\frac{1}{n}\mathbf{D}'\mathbf{P}_Z\mathbf{D} \xrightarrow{p} \mathbf{A} \succ 0, \quad \frac{1}{\sqrt{n}}\mathbf{D}'\mathbf{P}_Z\mathbf{u} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{S}),$$

so by Slutsky $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{A}^{-1}\mathbf{S}\mathbf{A}^{-1})$.

The robust variance estimator is the familiar sandwich:

$$\hat{V}(\hat{\beta}) = (\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}\mathbf{D}'\mathbf{P}_Z\hat{V}(\mathbf{u})\mathbf{P}_Z\mathbf{D}(\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1},$$

plugging in $\hat{V}(\mathbf{u}) = \text{diag}(\hat{u}_i^2)$, and clustering exactly as in Week 2. **Note the residuals $\hat{u}_i$ must be computed with the actual D_i , not $\hat{D}_i$** ; this is the other reason not to run the stages by hand.

LATE

Four Types of People (cont.)

With a binary instrument and binary treatment, write $D_i(z)$ for the treatment unit i would take if assigned $Z_i = z$. Every unit is one of four types:

Type	$D_i(0)$	$D_i(1)$	
Compliers	0	1	do what the instrument says
Always-takers	1	1	treated regardless
Never-takers	0	0	untreated regardless
Defiers	1	0	do the opposite

- We never observe anyone's type: we see $D_i(1)$ or $D_i(0)$, never both. Type is a latent variable, and this is the same fundamental problem as Week 5, one level up.

Four Types of People (cont.)

- Only **compliers** have their treatment moved by the instrument.
- **Monotonicity** assumes there are no defiers: the instrument pushes everyone in the same direction, or not at all.

Deriving the LATE: the Numerator

Write the reduced form using the switching equation $Y_i = Y_i(D_i(Z_i))$ and split by type. Because Z is independent of potential outcomes and types,

$$\begin{aligned}
 & E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0] \\
 &= \Pr(\text{AT}) \underbrace{(E[Y_i(1) \mid \text{AT}] - E[Y_i(1) \mid \text{AT}])}_{=0} \\
 &+ \Pr(\text{NT}) \underbrace{(E[Y_i(0) \mid \text{NT}] - E[Y_i(0) \mid \text{NT}])}_{=0} \\
 &+ \Pr(\text{C}) (E[Y_i(1) \mid \text{C}] - E[Y_i(0) \mid \text{C}]) \\
 &- \Pr(\text{D}) (E[Y_i(1) \mid \text{D}] - E[Y_i(0) \mid \text{D}]).
 \end{aligned}$$

Always- and never-takers take the *same* treatment under both values of Z , so **by exclusion** their outcomes are unchanged and they contribute zero. **Monotonicity** kills the last line by assuming $\Pr(\text{D}) = 0$, note that defiers would enter with the *wrong sign*.

Deriving the LATE: the Denominator

The first stage recovers the complier share:

$$\begin{aligned} E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0] &= \Pr(C) + \Pr(AT) - (\Pr(AT)) \\ &= \Pr(C). \end{aligned}$$

Dividing:

The LATE theorem (Imbens and Angrist 1994)

$$\frac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{E[D \mid Z = 1] - E[D \mid Z = 0]} = E[Y_i(1) - Y_i(0) \mid i \text{ is a complier}].$$

IV does not estimate the ATE. It estimates the average effect for the subpopulation whose treatment status the instrument actually changed, a group you cannot identify by name, defined by the instrument you happened to choose.

Living With the LATE

- If effects are homogeneous, $LATE = ATE$ and none of this matters. That is rarely a defensible assumption in sociology.
- Different instruments for the same treatment estimate *different* LATEs. When two IV papers disagree, that is not necessarily a contradiction; it may be two correct answers to two different questions.
- **You can describe the compliers even though you cannot name them:** their share is the first stage, and their covariate means are recoverable by reweighting (Abadie's κ).
- **Compliers are often unusual:** for quarter-of-birth they are marginal dropouts; for judge-leniency designs they are defendants near the decision margin. Whether that group is the policy-relevant one is a substantive question you must answer, not a technicality.

Always characterize your compliers. It is the difference between an interpretable LATE and a number.

Weak IV

Weak Instruments Are Worse Than No Instrument

Recall $\hat{\beta}_{IV} = \text{Cov}_n(Z, Y) / \text{Cov}_n(Z, D)$. If the denominator is near zero, small violations of exclusion get *divided by* a small number:

$$\text{plim } \hat{\beta}_{IV} = \beta + \frac{\text{Cov}(Z, \epsilon)}{\text{Cov}(Z, D)}.$$

- An instrument that is one-tenth as endogenous as D but one-hundredth as strongly correlated with it produces **more** bias than plain OLS.
- In matrix terms ★, the variance depends on $(\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}$. If the instruments explain little of $\mathbf{D}$ that matrix is small and its inverse is large.
- Worse, under weak identification $\hat{\beta}_{2SLS}$ has **no Gaussian limit**: the limiting distribution is non-normal and heavy-tailed, so the conventional t -test fails no matter how large n is.
- The problem *worsens* with more instruments, because many weak instruments overfit the first stage, the Bound–Jaeger–Baker point.

Diagnosing Instrument Strength

- **First-stage F** on the excluded instruments. The traditional rule of thumb is $F > 10$ (Staiger and Stock 1997; Stock and Yogo 2005), derived so that 2SLS bias is under 10% of OLS bias.
- **That rule is about bias, not about test size, and it is now known to be far too lax for inference.** Lee, McCrary, Moreira, and Porter (2022) show that a conventional 5% t -test has correct size only when the first-stage F exceeds roughly **104**. Their $\mathfrak{t}F$ procedure replaces the 1.96 critical value with a larger one read off F : at $F = 10$ the adjusted critical value is about **3.43**, so the honest interval is roughly **1.75** times wider than the printed one.
- Use a *robust* first-stage F if you use robust or clustered standard errors in the second stage.
- With several endogenous regressors, use the *Kleibergen–Paap* rank statistic; a strong F on each instrument separately is not enough.

Anderson–Rubin: Inference That Never Divides

The problem is the division by a noisy first stage. Anderson and Rubin (1949) avoid it by testing the moment condition directly.

To test $H_0 : \beta = \beta_0$, note that under the null $E[(Y_i - \beta_0 D_i) Z_i] = 0$. Form the statistic

$$C(\beta_0) = \frac{\left| \mathbb{E}_n[(Y_i - \beta_0 D_i) Z_i] \right|^2}{\text{Var}_n[(Y_i - \beta_0 D_i) Z_i]/n} \xrightarrow{d} \chi_1^2 \quad \text{under } H_0,$$

(with variables residualized on any exogenous controls first). Reject at level α if $C(\beta_0) > c_{1-\alpha}$, the $(1 - \alpha)$ quantile of χ_1^2 .

- Operationally: regress $Y_i - \beta_0 D_i$ on the instruments and test whether their coefficients are jointly zero.
- **The first stage never appears in the denominator**, so the test is valid whatever the instrument's strength.

Inverting the Test

A confidence region is the set of null values the data cannot reject:

$$CR(\beta) = \{\beta_0 \in \Theta : C(\beta_0) \leq c_{1-\alpha}\}.$$

In practice: grid over β_0 , compute $C(\beta_0)$, keep the values that survive.

- The region can be an interval, a *union* of disjoint intervals, the whole real line, or *empty*. None of these is a bug.
- An **unbounded** region is the honest statement that the data cannot pin β down, which is precisely what a conventional interval hides when the instrument is weak.
- An **empty** region can arise only when the model is *overidentified*, with one instrument and one endogenous regressor the 2SLS value always satisfies the moment exactly. When it does happen it is evidence against the overidentifying restrictions.
- `ivmodel`, `ivDiag`, or a hand-coded grid; it is ten lines of code, and we write them in lab.

Overidentification Tests: Read Them Carefully

With more instruments than endogenous regressors, the Sargan–Hansen J test asks whether the instruments agree with one another.

- **Rejection** says at least one instrument is invalid, *or* that effects are heterogeneous, so the instruments estimate different LATEs and are supposed to disagree. It does not say which.
- **Non-rejection is nearly worthless.** It is consistent with all the instruments being invalid *in the same direction*, and the test has little power.
- The J test cannot validate exclusion. Nothing can. The defence is substantive: tell a story about why Z could not affect Y except through D , and look for *placebo outcomes* it should not move and *placebo samples* in which the first stage should be absent.

Practice

A Reader's Checklist (cont.)

Adapted from Sovey and Green (2011). Any IV paper should report:

1. The instrument, and an explicit argument for exclusion, not just a citation to a prior paper that used it.
2. The **first stage**, its coefficient, and its robust F .
3. The **reduced form** (Y on Z), with its estimate and interval. It needs no instrument-strength assumption, so it is the most robust piece of evidence in the design, and if it is indistinguishable from zero the IV estimate rests on very little.
4. The **OLS** estimate alongside, so the reader can see the direction and size of the correction, and whether it goes the way theory predicts.
5. Who the **compliers** are, and their share.
6. Weak-instrument-robust inference if F is not comfortably large.
7. The just-identified estimate if the paper is overidentified.

A Reader's Checklist (cont.)

8. Placebo checks: outcomes and subgroups Z should not affect.

If a paper reports only the 2SLS coefficient, that is a red flag.

When IV Is the Wrong Tool (cont.)

- **No credible instrument exists.** This is the normal state of affairs. An implausible instrument is worse than an honest selection-on-observables analysis with sensitivity bounds.
- **The LATE is not the quantity of interest.** If your policy question concerns always-takers, the compliers' effect does not answer it.
- **Interpretation drifts.** Papers routinely estimate a LATE and then discuss it as if it were an ATE.
- **Monotonicity is doing hidden work.** In judge-leniency and examiner designs, it must hold for *every* judge simultaneously, a much stronger claim than it first appears.

When IV Is the Wrong Tool (cont.)

The design-based cousins

When the instrument is a discontinuity in an assignment rule, you have **RDD** (Week 9), and indeed fuzzy RD *is* local IV. When it is a policy change over time, you have **DID** (Week 10).

Looking Ahead

- IV replaces “I measured all confounders” with “I found variation that is as good as random.” Both are assumptions; the second is easier to argue about substantively, and easier for readers to attack, which is a virtue.
- The price is the estimand: LATE, for compliers you cannot name, defined by the instrument you chose.
- Notice how much identification work was done by *functional form* in AJR, “linear in distance from the equator, plus continent dummies.” Week 13 asks briefly what can be done when you are unwilling to assume that.

Midterm

Thursday, October 15, in class, covering Weeks 1–7. Open book and open note, timed, no coding. Tuesday October 13 is Fall break.

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